

Physics-Based Cutting Force Model for CNC Machining

Mechanical Engineering Plan Document

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1 Objective

Develop a physics-based model to predict cutting forces in CNC machining based on:

- Material properties
- Tool geometry
- Cutting parameters (speed, feed, depth)

2 Theoretical Framework

2.1 Cutting Force Model

$$F = K_c \cdot A_c \quad (1)$$

2.2 Chip Thickness

$$h(\phi) = f_z \sin(\phi) \quad (2)$$

2.3 Milling Force Components

$$F_t = K_t \cdot b \cdot h \quad (3)$$

$$F_r = K_r \cdot b \cdot h \quad (4)$$

2.4 Material Dependence

$$K_c = K_{c1.1} \cdot h^{-m} \quad (5)$$

2.5 Corrected Speed Relations

Imperial:

$$RPM = \frac{SFM \cdot 3.82}{D} \quad (6)$$

SI:

$$N = \frac{1000 \cdot V}{\pi D} \quad (7)$$

2.6 Feed Rate

$$F = N \cdot Z \cdot f_z \quad (8)$$

3 Material Parameters (From Shop Data)

Material	SFM Range	Typical K_c (MPa)
Aluminum (6061)	800–1200	600
Aluminum (cast)	600–900	700
Mild Steel (1018)	300–500	1800
Alloy Steel (4140)	200–400	2200
Stainless Steel (304)	150–300	2400
Tool Steel (D2)	100–250	3000
Cast Iron	400–700	1600
Brass	500–1000	500
Titanium (Ti-6Al-4V)	80–200	3000
Nickel Alloys (Inconel)	50–150	3500

Table 1: Cutting parameters derived from shop-floor tables [1]

4 Practical Interpretation

- Feed controls chip thickness and force
- Speed controls temperature
- Depth controls total load

Key observation:

If chips are dust $\rightarrow$ feed too low

If chatter occurs $\rightarrow$ reduce feed first

(Shop-floor rule)

5 Simplified Force Model

$$F \approx K_c \cdot f_z \cdot a_p \cdot a_e \quad (9)$$

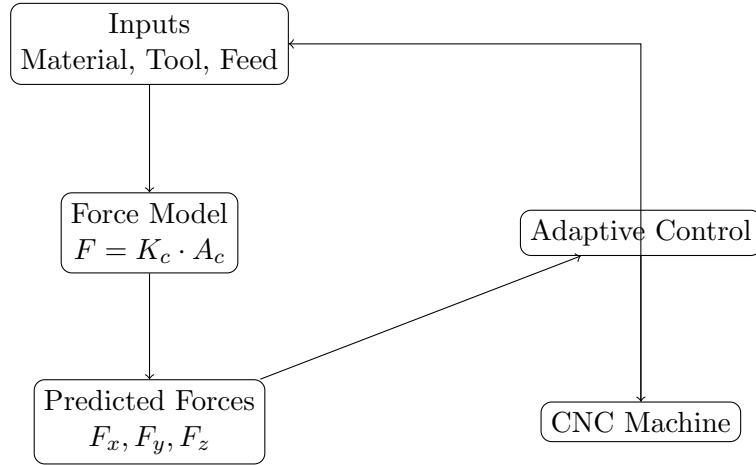
6 Implementation Plan

6.1 Phases

1. Analytical model development
2. Data collection (spindle load, cuts)
3. Parameter identification

4. Real-time CNC integration
5. Adaptive optimization

7 System Flow Diagram



8 Validation Plan

- Compare predicted vs measured forces
- Evaluate tool wear and chatter
- Validate across materials and operations

9 Conclusion

Cutting force is primarily governed by chip formation physics. A successful model integrates:

- Mechanistic equations
- Empirical shop-floor adjustments